Rotation

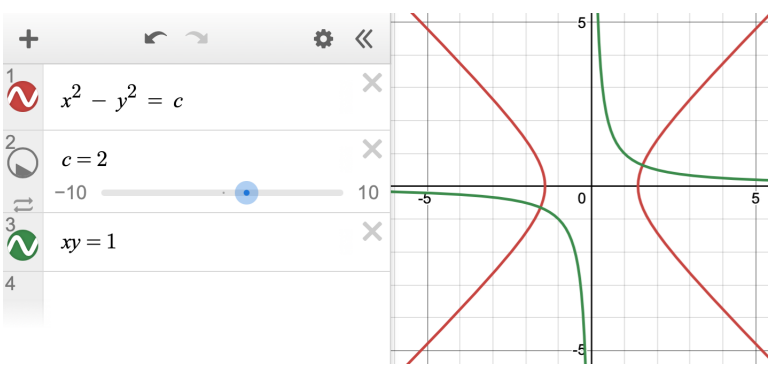
introduction

In thinking about conic sections they are often presented in *standard* orientation. For example $y = ax^2$ or $x^2 + y^2 = r^2$ and also

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \qquad \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

which are the equations of a parabola, a circle, an ellipse, and a hyperbola, respectively. Each is centered at the origin.

The parabola opens up. If $a > b$ the ellipse has its long dimension along the x-axis. The hyperbola (with $a = b = c$) is as shown in the figure as the red curve.

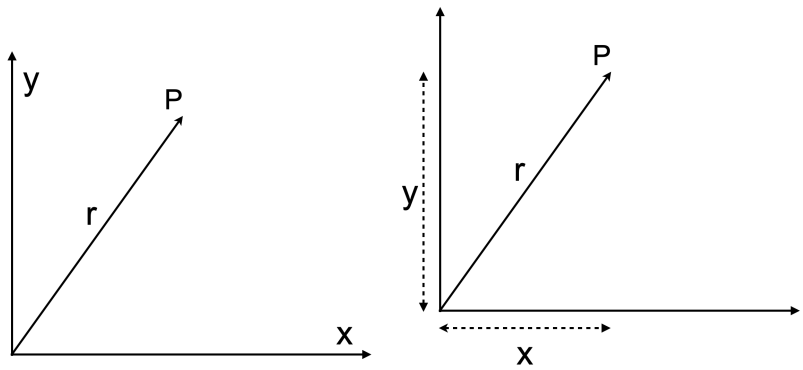


But that hyperbola can be *rotated*. In fact, the familiar equation $xy = 1$ is the same shape, rotated by 45° (green curve above). The point of closest approach to the center is the same for each $\sqrt{2}$, and the axes are perpendicular — a consequence of having $a = b = c$.

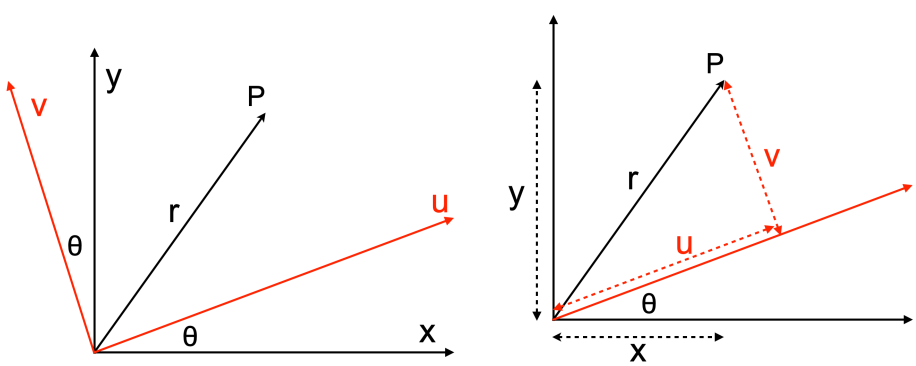
We want to understand how to transform equations into their rotated versions.

basic idea

Start with the standard picture from analytical geometry, courtesy of Fermat and Descartes. Pick a point for the origin. Draw perpendicular x- and y-axes (the fancy word is orthogonal).



The point P lies a distance r from the origin. In the xy-coordinate system, $P = (x, y)$ and $r^2 = x^2 + y^2$.



Now consider a second coordinate system with u- and v-axes, rotated counter-clockwise by an angle θ. In the new system, $P = (u, v)$.

We will find a formula to give u as a function of x and y (and also θ),

abbreviated $u(x, y, \theta)$. Similarly, we will get $v(x, y, \theta)$.

We will also get $x(u, v, \theta)$ and $y(u, v, \theta)$. This can be understood as rotation of the coordinate system CW.

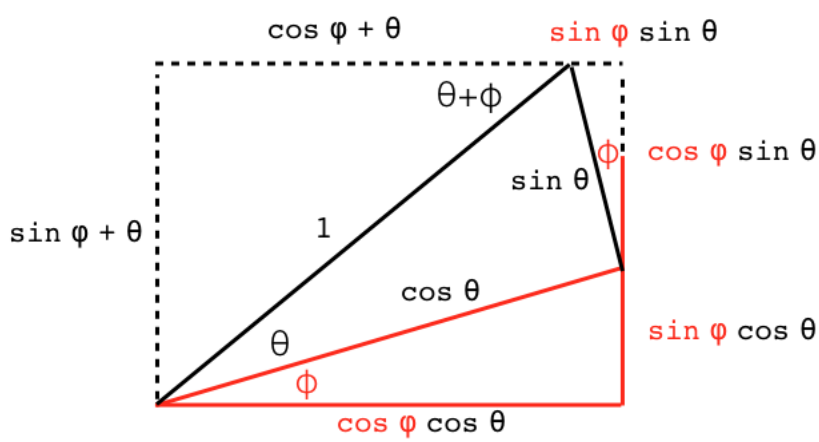
This can be done algebraically, but we do it in other ways as well. In fact, the first approach (a visual derivation) gives the formulas going from u, v to x, y.

Before we get into the specifics, note that a rotation of the coordinate system CCW, from x, y-axes to u, v-axes, can equally be viewed as the rotation of the point P, or a vector from the origin to P in the opposite, CW direction.

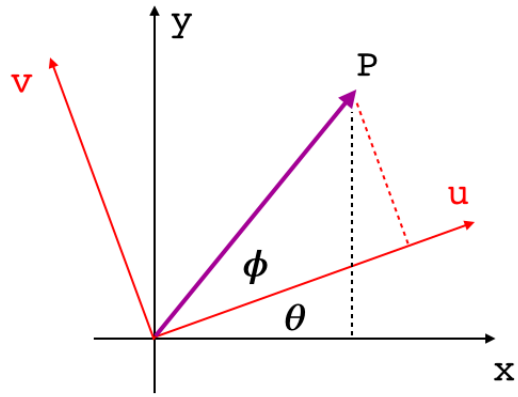
We will adopt the convention that when we rotate the coordinate system CCW, the operation is carried out by an operator R_{ccw}. We'll define R below. Here we simply observe that the R_{ccw} operator rotates points and vectors CW, as strange as that sounds.

Stewart

Probably the easiest method I have found, in Stewart, depends on knowing the sum of angles formulas.



It goes like this.



Let the point P lie on the unit circle, with $r = 1$. The coordinates of point P in the u, v system are naturally expressed in terms of ϕ:

$$u = \cos\phi$$

$$v = \sin\phi$$

while x and y are naturally expressed in terms of the combined angle $\theta + \phi$.

$$x = \cos(\theta + \phi)$$

$$y = \sin(\theta + \phi)$$

Now, use the sum formula for cosine:

$$x = \cos\theta\cos\phi - \sin\theta\sin\phi$$

Plug in from above:

$$x = u\cos\theta - v\sin\theta$$

It's as easy as that.

In the same way:

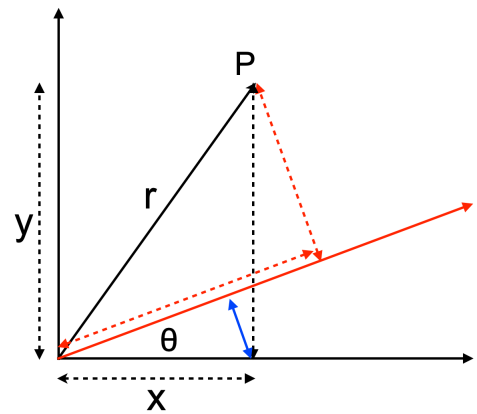
$$y = \sin(\theta + \phi)$$

$$= \sin\theta\cos\phi + \cos\theta\sin\phi$$

$$= u \sin\theta + v \cos\theta$$

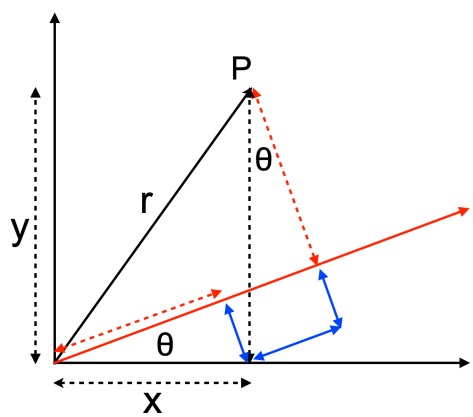
We have x, y in terms of u, v. So this derivation naturally gives R_{cw}.

drawing rectangles



Next we look at two methods that work by drawing rectangles that help us to see the pieces.

The key step of the first approach is to draw a line that *makes x the hypotenuse* of a right triangle. That's the blue line in the figure above.

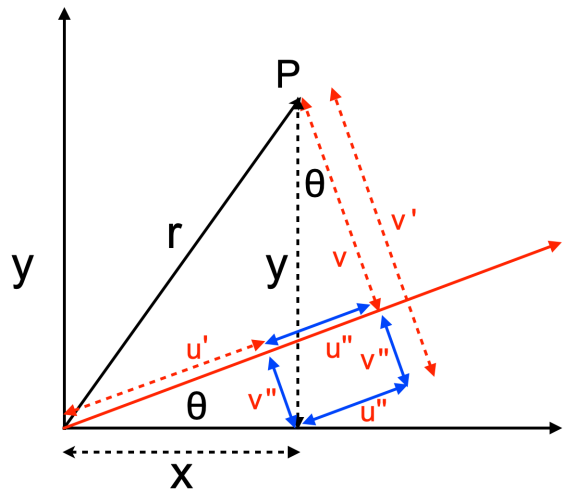


Then, draw the adjacent side of the rectangle to make y the hypotenuse of a right triangle.

θ, the angle between x and u axes, is also the angle between y and v axes. Note the angle near P in the diagram.

Basic trigonometry gives:

$$\sin\theta = \frac{v''}{x} = \frac{u''}{y} \qquad \cos\theta = \frac{u'}{x} = \frac{v'}{y}$$



$$u' = x\cos\theta \qquad u'' = y\sin\theta$$

$$v' = y\cos\theta \qquad v'' = x\sin\theta$$

We can therefore write u as the sum of two segments:

$$u = u' + u'' = x\cos\theta + y\sin\theta$$

And write v as the difference of two segments.

$$v = v' - v'' = y\cos\theta - x\sin\theta$$

matrix form

A convenient notation is to put them in a matrix that multiplies a vector:

$$\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$$

This is R_{ccw}, the matrix for counter-clockwise rotation of coordinates, from x, y-axes to u, v-axes.

We can look at this transformation in a different way, namely that this is a clockwise rotation of the point P.

We can also write R_{cw}:

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

It's just a matter of moving a minus sign. In vector terms, the operation is called a transpose.

algebraic manipulation

We will find two functions that turn u, v into x, y. It's easy when you know how. We have:

$$u = x\cos\theta + y\sin\theta$$
$$v = -x\sin\theta + y\cos\theta$$

To solve for x we need to lose y.

Multiply the first equation by $\cos\theta$ and the second by $-\sin\theta$ and add. The terms with y in them cancel.

$$u\cos\theta - v\sin\theta = x(\cos^2\theta + \sin^2\theta) = x$$

Also, multiply the first equation by $\sin\theta$ and the second by $\cos\theta$ and add:

$$u\sin\theta + v\cos\theta = y(\cos^2\theta + \sin^2\theta) = y$$

In matrix form, this is R_{cw}, which we gave above.

If you know about matrices, you will realize that the two matrices must be inverses, since rotating first one direction and then the other, by the same angle, leaves the point unchanged.

$$R_{ccw}\ R_{cw} = I$$

To invert a 2×2 matrix, transpose it and divide by the determinant (which is 1 here). So

$$\begin{aligned} R_{ccw} &= \frac{1}{|R_{cw}|} \cdot R_{cw}^T = R_{cw}^T \\ &= \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \end{aligned}$$

We can also check to make sure that these work correctly with the unit vectors. Recall that rotation of a point CCW is the same as rotation of the coordinate system CW, so we use R_{cw}:

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix}$$

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}$$

The rotated unit vector $\hat{\mathbf{j}}$ lies in the second quadrant.

another view of CCW and CW rotation

Given $x = f(x, y)$ and $y = g(x, y)$, we wish to convert these to $u = f'(x, y)$ and $v = g'(x, y)$

One way is to switch symbols x, y for u, v, and at the same time, substitute $-\theta$ for θ. What this amounts to is relabeling our diagram with x, y being the rotated axes, and then rotating in the opposite direction (cw rather than ccw).

$$x = u \cos\theta - v \sin\theta$$

switch

$$\begin{aligned} u &= x \cos -\theta - y \sin -\theta \\ &= x \cos\theta + y \sin\theta \end{aligned}$$

(Recall that $\cos -x = \cos x$ and $\sin -x = -\sin x$).

For y

$$y = u \sin\theta + v \cos\theta$$

switch

$$\begin{aligned} v &= x \sin -\theta + y \cos -\theta \\ v &= -x \sin\theta + y \cos\theta \end{aligned}$$

vectors

Another approach is to guess. Start from vectors and matrices and remember that we have a 2×2 matrix.

Suppose the angle is $\theta = 0$. Then we want the matrix to be the identity matrix

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

That naturally means that $\cos\theta$ is for entries along the diagonals and $\sin\theta$ at the other positions.

If the angle is $90°$ we want to rotate the unit vector $\hat{\mathbf{i}}$, to become unit vector $\hat{\mathbf{j}}$. That means we want

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Putting it all together

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

For $\theta = 90°$:

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \qquad \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

That looks good. So

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} u \\ v \end{bmatrix}$$

means that

$$u = x\cos\theta - y\sin\theta$$
$$v = x\sin\theta + y\cos\theta$$

Rotating the points counter-clockwise is equivalent to rotating the coordinate system clockwise.

rotation by $+\theta$ then $-\theta$

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

another way

We can look at this in a different way. Write

$$\begin{bmatrix} x \\ y \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

In this representation, the vector $\langle x, y \rangle$ is a linear combination of the unit vectors $\hat{\mathbf{i}} = \langle 1, 0 \rangle$ and $\hat{\mathbf{j}} = \langle 0, 1 \rangle$.

To rotate the point, we just want to use a different set of unit vectors. The new unit vectors (for the rotated axes) are $\langle \cos\theta, \sin\theta \rangle$ and $\langle -\sin\theta, \cos\theta \rangle$.

If you compute their lengths, it is clear that they are, in fact, unit vectors.

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = x \begin{bmatrix} \cos\theta \\ \sin\theta \end{bmatrix} + y \begin{bmatrix} -\sin\theta \\ \cos\theta \end{bmatrix}$$

Written as a matrix multiplication, this is

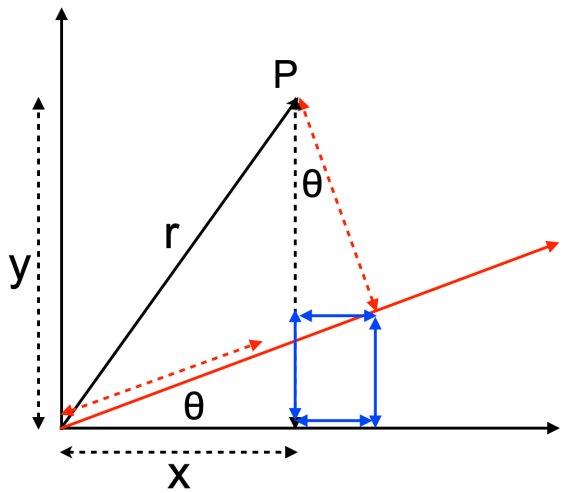
$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$$

another rectangle picture

At the risk of confusing you, there is another rectangle picture which gives $x(u, v, \theta)$ and $y(u, v, \theta)$.

Recall that previously we drew a rectangle whose sides made x and y the hypotenuse of two different right triangles.

The new approach makes both u and v the hypotenuse of two different right triangles.



We leave it as an exercise to derive $x(u, v, \theta)$ and $y(u, v, \theta)$. These equations have already been introduced above.

examples: parabola

Consider $y = x^2$ rotated through an angle of 45°.

We need expressions for x and y in terms of u and v. Let us just try R_{cw} and see where we get.

$$\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix} = \begin{bmatrix} x \\ y \end{bmatrix}$$

$$x = u\cos\theta - v\sin\theta$$

$$v = u\sin\theta + v\cos\theta$$

Furthermore, $\cos\theta = \sin\theta = 1/\sqrt{2}$.

$$x = 1/\sqrt{2}(u - v)$$

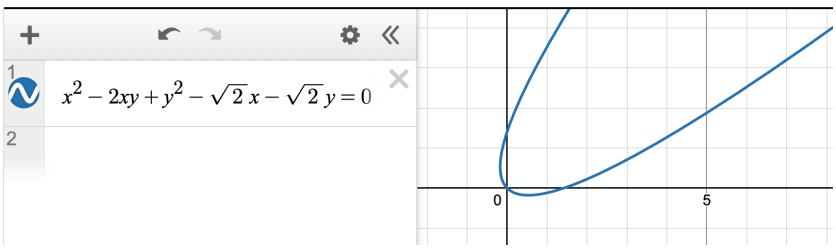
$$y = 1/\sqrt{2}(u + v)$$

so

$$y = x^2$$
$$1/\sqrt{2} \cdot (u+v) = (\frac{1}{\sqrt{2}})^2 \cdot (u-v)^2$$
$$\sqrt{2}(u+v) = (u-v)^2 = u^2 - 2uv + v^2$$
$$u^2 - 2uv + v^2 - \sqrt{2}u - \sqrt{2}v = 0$$

Desmos doesn't like u and v so just for the graph, we must substitute x and y:

$$x^2 - 2xy + y^2 - \sqrt{2}x - \sqrt{2}v = 0$$



We've got a parabola rotated 45° CW. Its center is still at the origin.

We said that R_{cw} rotates the coordinate system CW but vectors and points in the opposite direction. What happened? Notice that u, v were on the left side of the matrix equation above. In other words, the parabola in the panel above was rotated by 45° CCW to give $y = x^2$.

examples: hyperbola

The standard equation of a hyperbola is usually given as

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

Yet the simplest hyperbola is $xy = 1$. What's the explanation? We gave part of the answer in the introduction. Here is a more complete explanation.

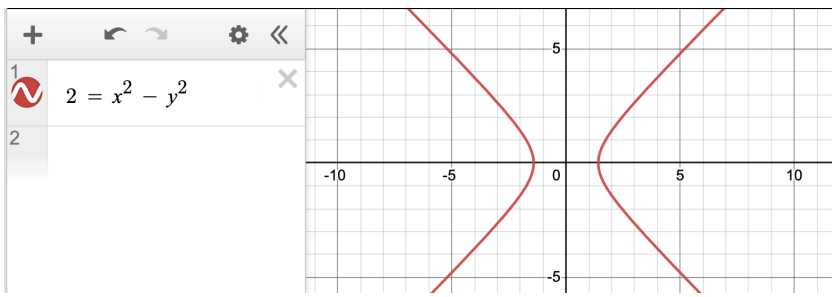
Going back to our first result:

$$x = u\cos\theta - v\sin\theta$$

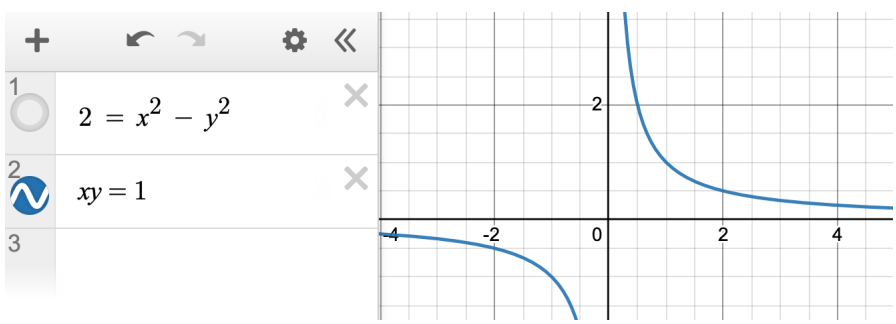
$$v = u\sin\theta + v\cos\theta$$

$$(u\cos\theta - v\sin\theta)(u\sin\theta + v\cos\theta)$$
$$\frac{1}{2}\cdot(u-v)(u+v)$$
$$2 = (u^2 - v^2)$$

Again, switch to x, y for Desmos:



When this guy is rotated by 45 CCW, it becomes the familiar graph of $xy = 1$.



The R_{cw} matrix operates on u, v, the first form, rotating the *points* ccw to give the familiar $xy = 1$.

finding the angle

Suppose we are given a conic section like:

$$Ax^2 + Bxy + Cy^2 + Dx + Ey + F = 0$$

and we wish to rotate it to the standard orientation.

Take the equations for R_{cw}:

$$x = u\cos\theta - v\sin\theta$$

$$y = u\sin\theta + v\cos\theta$$

and substitute. As a preliminary

$$x^2 = u^2\cos^2\theta - 2uv\sin\theta\cos\theta + v^2\sin^2\theta$$

$$xy = u^2\sin\theta\cos\theta + uv\cos^2\theta - uv\sin^2\theta - v^2\sin\theta\cos\theta$$

$$y^2 = u^2\sin^2\theta + 2uv\sin\theta\cos\theta + v^2\cos^2\theta$$

We only need to consider the terms that have xy in them. Remembering to pick up A, B, C we have for the cofactors of xy:

$$-2A\sin\theta\cos\theta + B\cos^2\theta - B\sin^2\theta + 2C\sin\theta\cos\theta$$

What is needed is to make this term zero.

$$(C - A)(2\sin\theta\cos\theta) + B(\cos^2\theta - \sin^2\theta) = 0$$

The sum of angles formulas come in handy now:

$$(C - A)(\sin 2\theta) + B(\cos 2\theta) = 0$$

$$\cot 2\theta = \frac{A - C}{B}$$

The hyperbola $xy = 1$ has both A and C equal to zero, while B is just 1. What angle has a cotangent of zero? The one whose cosine is zero, namely $\pi/2$ or $90°$. That is 2θ, so our final answer is $45°$.

For the other example

$$x^2 - 2xy + y^2 - \sqrt{2}x - \sqrt{2}v = 0$$

First note that $B^2 = 4AC$ so this passes the parabola test. Then, again we need $\cot 2\theta = 0$, with the same angle as the final answer.